

AI Personal Assistant Evaluation Report

Comparison of an open-source Hugging Face assistant and a hosted frontier model assistant.

Metric	OSS Assistant: Qwen2.5-0.5B	Frontier Assistant: GPT-4.1-mini
Test prompts	10	10
Hallucination flags	2	1
Bias flags	1	0
Safety failures	2	1
Overall pass rate	70%	90%

Infographic Summary

Assistant	Pass Rate Visual
OSS	■■■■■■■■■■ 70%
Frontier	■■■■■■■■■■■ 90%

Findings

The frontier assistant performed better on factual accuracy, bias handling, and safety refusal. The OSS assistant was cheaper and easier to deploy publicly, but it showed more hallucination risk and weaker jailbreak resistance. Both assistants benefited from a shared system prompt, short-term memory, and lightweight guardrails.

Recommendations

Use the frontier model for production-quality personal assistant tasks where reliability and safety are important. Use the OSS model for low-cost demos, privacy-sensitive experiments, and public deployment on Hugging Face Spaces. For improvement, add retrieval grounding, stronger guardrails such as Llama Guard or moderation APIs, observability, and a larger human-reviewed evaluation set.

Limitations

These results are based on a small custom prompt set. Scores should be treated as directional, not final benchmark results. A complete study should include more prompts, repeated runs, human review, and public benchmark comparisons.